Experiment ledger and glossary

The observed positive KID difference contradicts the proposed oscillator improvement in this regime.

All networks generate unconditional images in pixel space. They share independent conditional flow matching, AdamW, continuous time conditioning, 4×4 patches, and an EMA sampling checkpoint. MNIST is development data; CIFAR-10 is the final comparison domain. There is no curriculum or augmentation. Classification accuracy, text fill rate, and task-completion accuracy are not applicable; run completion is reported separately from image quality.

Network	Feature computation	Training objective	Role
onn	Coupled 4D unit-vector rotations; four recurrent blocks	Velocity MSE	Proposed generator
dit	Four time-conditioned attention/MLP blocks	Velocity MSE	Parameter-matched transformer
conv	Convolution + SiLU + GroupNorm recurrence	Velocity MSE	Conventional recurrent control
no_coupling	Oscillators with coupling convolution removed	Velocity MSE	Trained removal; lower capacity
no_frequency	Oscillators with natural-frequency term removed	Velocity MSE	Trained removal; slightly lower capacity

Training outcomes

Run	Status	Parameters	Updates	Train seconds	Images/s	Validation MSE
checks/cifar10-metric-smoke	untrained	1297712	0	0	—	—
checks/mnist-smoke	step_limit	1289488	100	33.9622	376.889	2.00869
study/ablations/cifar10-no_coupling-s101	completed	707888	9405	2700.28	445.82	0.262754
study/ablations/cifar10-no_coupling-s102	completed	707888	10045	2700.04	476.2	0.266352
study/ablations/cifar10-no_frequency-s101	completed	1295664	10320	2700.08	489.23	0.211623
study/ablations/cifar10-no_frequency-s102	completed	1295664	10583	2700.13	501.688	0.216272

Run	Status	Parameters	Updates	Train seconds	Images/s	Validation MSE
study/controls/untrained-conv	untrained	1296688	0	0	—	—
study/controls/untrained-dit	untrained	1298864	0	0	—	—
study/controls/untrained-onn	untrained	1297712	0	0	—	—
study/main/cifar10-conv-s101	completed	1296688	12473	2700.18	591.272	0.203593
study/main/cifar10-conv-s102	completed	1296688	12365	2700.18	586.153	0.210089
study/main/cifar10-conv-s103	completed	1296688	12482	2700.07	591.725	0.212733
study/main/cifar10-conv-s104	completed	1296688	12480	2700.04	591.637	0.210087
study/main/cifar10-conv-s105	completed	1296688	12236	2700.17	580.04	0.213325
study/main/cifar10-conv-s106	completed	1296688	11489	2700.03	544.658	0.221481
study/main/cifar10-dit-s101	completed	1298864	16959	2700.17	803.93	0.192738
study/main/cifar10-dit-s102	completed	1298864	17303	2700.06	820.271	0.197029
study/main/cifar10-dit-s103	completed	1298864	17148	2700.12	812.907	0.201229
study/main/cifar10-dit-s104	completed	1298864	16981	2700.03	805.017	0.198842
study/main/cifar10-dit-s105	completed	1298864	15926	2700.13	754.973	0.201822
study/main/cifar10-dit-s106	completed	1298864	15345	2700.11	727.438	0.208719
study/main/cifar10-onn-s101	completed	1297712	8589	2700.05	407.175	0.215884

Run	Status	Parameters	Updates	Train seconds	Images/s	Validation MSE
study/main/cifar10-onn-s102	completed	1297712	8588	2700.09	407.122	0.220631
study/main/cifar10-onn-s103	completed	1297712	8632	2700.27	409.179	0.224278
study/main/cifar10-onn-s104	completed	1297712	8536	2700.11	404.654	0.221115
study/main/cifar10-onn-s105	completed	1297712	8248	2700.25	390.981	0.226086
study/main/cifar10-onn-s106	completed	1297712	7642	2700.3	362.248	0.234817
study/pilot/mnist-conv-s101	completed	1288464	5245	1200.05	559.445	0.215197
study/pilot/mnist-dit-s101	completed	1290640	6778	1200.04	722.965	0.203531
study/pilot/mnist-onn-s101	completed	1289488	3736	1200.25	398.425	0.231511

Distribution and cost measurements

Evaluation	Generated/real N	NFE	KID mean	KID subset SD	FID	Pixel SWD	ms/image
checks/cifar10-metric-smoke/evaluations/validation	128/5000	32	0.609741	0.0059603	487.666	—	21.3627
checks/mnist-smoke/validation-check	128/5000	32	—	—	—	1.10101	20.9486
study/ablations/cifar10-no_coupling-s101/evaluations/test-nfe32	5000/10000	32	0.26963	0.00267395	269.031	—	19.4271
study/ablations/cifar10-no_coupling-s102/evaluations/test-nfe32	5000/10000	32	0.267981	0.00290885	266.878	—	22.0612

Evaluation	Generated/ real N	NFE	KID mean	KID subset SD	FID	Pixel SWD	ms/image
study/ablations/cifar10-no_frequency-s101/evaluations/test-nfe32	5000/10000	32	0.169792	0.0025135	163.113	—	16.5046
study/ablations/cifar10-no_frequency-s102/evaluations/test-nfe32	5000/10000	32	0.160263	0.0024495	156.719	—	16.0385
study/controls/untrained-conv/evaluations/test	5000/10000	32	0.604058	0.00279767	456.73	—	19.9959
study/controls/untrained-dit/evaluations/test	5000/10000	32	0.601732	0.00281394	455.442	—	10.97
study/controls/untrained-onn/evaluations/test	5000/10000	32	0.599606	0.00259296	455.38	—	28.3087
study/main/cifar10-conv-s101/evaluations/test-nfe16	5000/10000	16	0.129412	0.0022568	127.666	—	7.59598
study/main/cifar10-conv-s101/evaluations/test-nfe32	5000/10000	32	0.132659	0.00224372	132.002	—	13.9111
study/main/cifar10-conv-s101/evaluations/test-nfe64	5000/10000	64	0.135467	0.00227747	135.195	—	28.4834
study/main/cifar10-conv-s101/evaluations/test-nfe8	5000/10000	8	0.127097	0.00227732	122.565	—	3.69969
study/main/cifar10-conv-s102/	5000/10000	32	0.137173	0.0022348	136.479	—	14.257

Evaluation	Generated/ real N	NFE	KID mean	KID subset SD	FID	Pixel SWD	ms/image
evaluations/test-nfe32							
study/main/cifar10-conv-s103/evaluations/test-nfe32	5000/10000	32	0.131977	0.00215866	130.711	—	17.6869
study/main/cifar10-conv-s104/evaluations/test-nfe32	5000/10000	32	0.131518	0.00212369	130.908	—	14.792
study/main/cifar10-conv-s105/evaluations/test-nfe32	5000/10000	32	0.13401	0.00216785	133.047	—	17.3853
study/main/cifar10-conv-s106/evaluations/test-nfe32	5000/10000	32	0.141894	0.00228545	141.231	—	16.7635
study/main/cifar10-dit-s101/evaluations/test-nfe16	5000/10000	16	0.0857987	0.00165292	98.6546	—	5.21638
study/main/cifar10-dit-s101/evaluations/test-nfe32	5000/10000	32	0.0909215	0.00172645	104.111	—	10.1586
study/main/cifar10-dit-s101/evaluations/test-nfe64	5000/10000	64	0.0952455	0.00181027	108.063	—	21.3938
study/main/cifar10-dit-s101/evaluations/test-nfe8	5000/10000	8	0.0838022	0.00163955	95.6179	—	2.4516
study/main/cifar10-dit-s102/evaluations/test-nfe32	5000/10000	32	0.0899517	0.00169648	103.125	—	9.77963

Evaluation	Generated/ real N	NFE	KID mean	KID subset SD	FID	Pixel SWD	ms/image
study/main/ cifar10-dit-s103/evaluations/test-nfe32	5000/10000	32	0.0920323	0.00184124	105.047	—	10.1741
study/main/ cifar10-dit-s104/evaluations/test-nfe32	5000/10000	32	0.0926866	0.00191923	105.141	—	13.4588
study/main/ cifar10-dit-s105/evaluations/test-nfe32	5000/10000	32	0.0905512	0.00181148	104.204	—	12.1196
study/main/ cifar10-dit-s106/evaluations/test-nfe32	5000/10000	32	0.0945936	0.00188419	106.249	—	9.76628
study/main/ cifar10-onn-s101/ evaluations/identity	5000/10000	32	0.188511	0.00274027	176.705	—	23.2444
study/main/ cifar10-onn-s101/ evaluations/shuffle	5000/10000	32	0.263509	0.00299769	240.955	—	23.6636
study/main/ cifar10-onn-s101/ evaluations/test-nfe16	5000/10000	16	0.178735	0.00259012	168.147	—	11.3092
study/main/ cifar10-onn-s101/ evaluations/test-nfe32	5000/10000	32	0.188511	0.00274027	176.705	—	21.099
study/main/ cifar10-onn-s101/ evaluations/test-nfe64	5000/10000	64	0.19384	0.00281791	181.258	—	46.2968
study/main/ cifar10-onn-s101/ evalua-	5000/10000	8	0.174378	0.00246556	162.823	—	6.22954

Evaluation	Generated/ real N	NFE	KID mean	KID subset SD	FID	Pixel SWD	ms/image
tions/test-nfe8							
study/main/cifar10-onn-s102/evaluations/identity	5000/10000	32	0.179221	0.00275456	170.893	—	26.7271
study/main/cifar10-onn-s102/evaluations/shuffle	5000/10000	32	0.257773	0.00364871	233.55	—	27.1896
study/main/cifar10-onn-s102/evaluations/test-nfe32	5000/10000	32	0.179221	0.00275456	170.893	—	21.6993
study/main/cifar10-onn-s103/evaluations/test-nfe32	5000/10000	32	0.181497	0.00273838	171.989	—	24.4268
study/main/cifar10-onn-s104/evaluations/test-nfe32	5000/10000	32	0.176938	0.00259902	167.294	—	24.6136
study/main/cifar10-onn-s105/evaluations/test-nfe32	5000/10000	32	0.186201	0.00282784	176.039	—	27.2132
study/main/cifar10-onn-s106/evaluations/test-nfe32	5000/10000	32	0.201672	0.0028731	187.592	—	21.3573
study/pilot/mnist-conv-s101/evaluations/validation	512/5000	32	—	—	—	0.063441	13.336
study/pilot/mnist-dit-s101/evaluations/validation	512/5000	32	—	—	—	0.0617331	9.42476

Evaluation	Generated/ real N	NFE	KID mean	KID subset SD	FID	Pixel SWD	ms/image
study/pilot/ mnist-onn-s101/evaluations/validation	512/5000	32	—	—	—	0.0714237	27.8529

Completeness

Recorded diagnostics: 1. Recorded failed attempts: 2. Completed primary pairs: 6/6. Missing values mean unmeasured, not zero. Individual records and PDFs are in `experiments/`; raw records remain in the run directories and published artifact manifests.

Glossary

Term	Meaning
Flow matching	Learn a velocity that transports Gaussian noise toward the image distribution.
Oscillator	A hidden vector whose direction changes under natural rotation and learned interactions.
Coupling	A learned convolution that lets hidden oscillators influence one another.
Flow time / internal time	Noise-to-image position / recurrent computation inside one velocity prediction.
NFE	Number of velocity-network evaluations used to generate each image.
KID	Unbiased squared kernel discrepancy between fixed Inception features; smaller is better and estimates may be negative.
KID subset SD	Variation across random metric subsets of one fixed pair of image populations, not across trained models.
FID	Gaussian feature-distribution distance; finite-sample biased, so sample count must accompany it.
Pixel SWD	Wasserstein distance averaged over random pixel-space projections; a MNIST diagnostic, not semantic accuracy.
Seed pair	Two independently initialized architectures using paired data/noise/time stream prefixes.
Confidence interval	A model-based repeated-study procedure; here an approximate t interval across six seed differences.
Sign-flip test	Enumerate all signs of paired differences; exact only under the specified exchangeability/symmetry null.
EMA	Exponential moving average of trainable weights, used for all reported samples.
Ablation / intervention	Retrain with a mechanism removed / perturb a trained model during inference.
Power	Probability of detecting an assumed effect under a planning model; six pairs have about 24% power at effect 0.8.